Project 1: xv6 Custom System Calls

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Topic: Inter-Process Communication (IPC) using Pipes
Instructor: Dr. Jaishree Mayank **Subject:** OS Lab – NSCS210/CSC211

What is a Pipe?

A pipe is a system call that allows two processes to communicate with each other. One process writes data into the pipe and another process reads from it.

Files Added

1. `user/pipetest.c` – Basic pipe communication
2. `user/pipetest2.c` – Bidirectional pipe
3. `user/pipetest3.c` – Pipe with error handling

Program 1: Basic Pipe (`pipetest.c`)

What it does:

- Creates a pipe between parent and child process
- Child writes `"Hello via pipe!"` to pipe
- Parent reads and prints the message

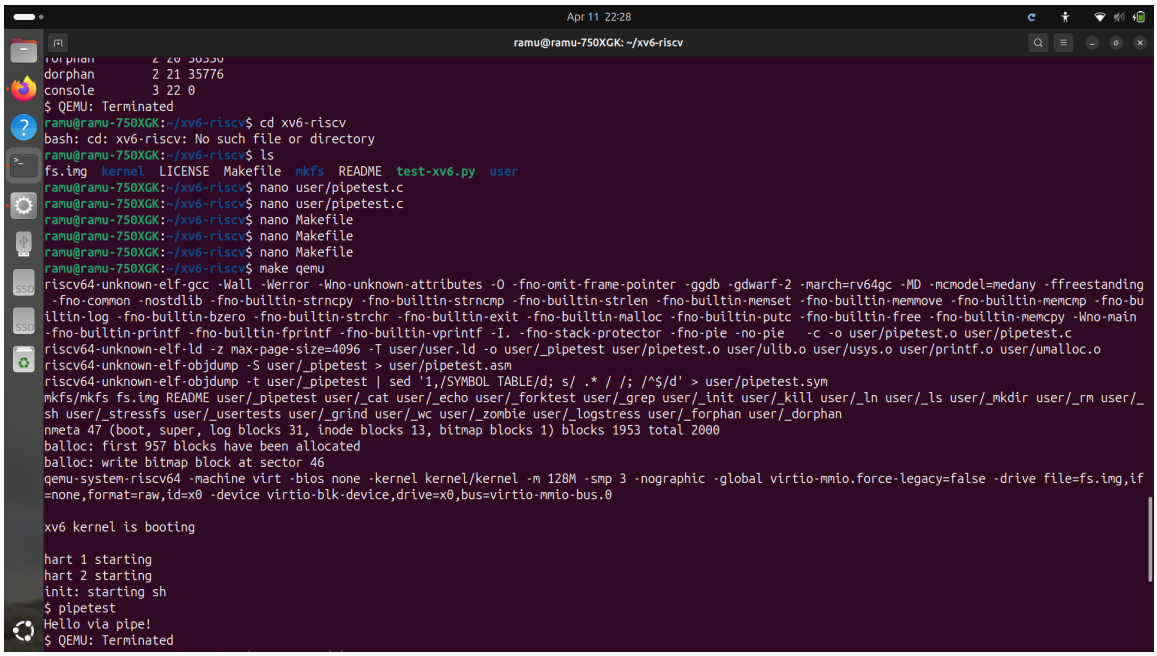
How to run:

```
$ pipetest
```

Output:

```
Hello via pipe!
```

Screenshot:



Program 2: Bidirectional Pipe (`pipetest2.c`)

What it does:

- Creates TWO pipes for two-way communication
- Parent sends message to child
- Child sends message back to parent
- Both processes communicate with each other

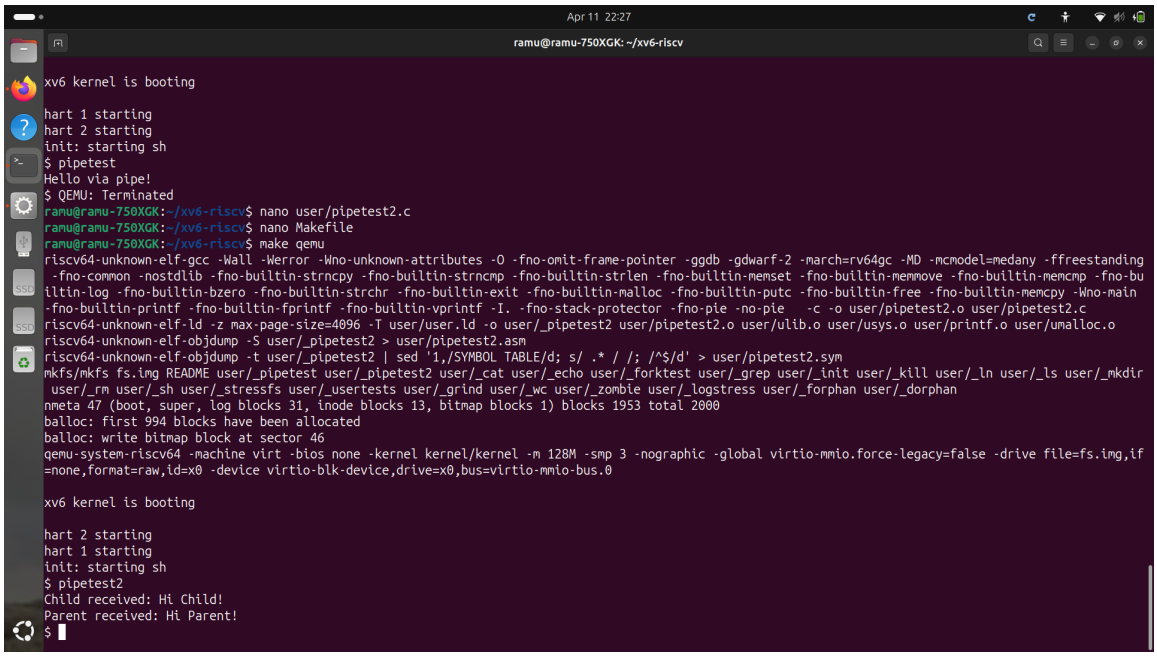
How to run:

```
$ pipetest2
```

Output:

```
Child received: Hi Child!
Parent received: Hi Parent!
```

Screenshot:



Program 3: Pipe with Error Handling (`pipetest3.c`)

What it does:

- Creates pipe with proper error checking
- Checks if pipe creation succeeds
- Checks if fork succeeds
- Checks if read/write succeeds
- Reports number of bytes transferred
- Closes pipe safely

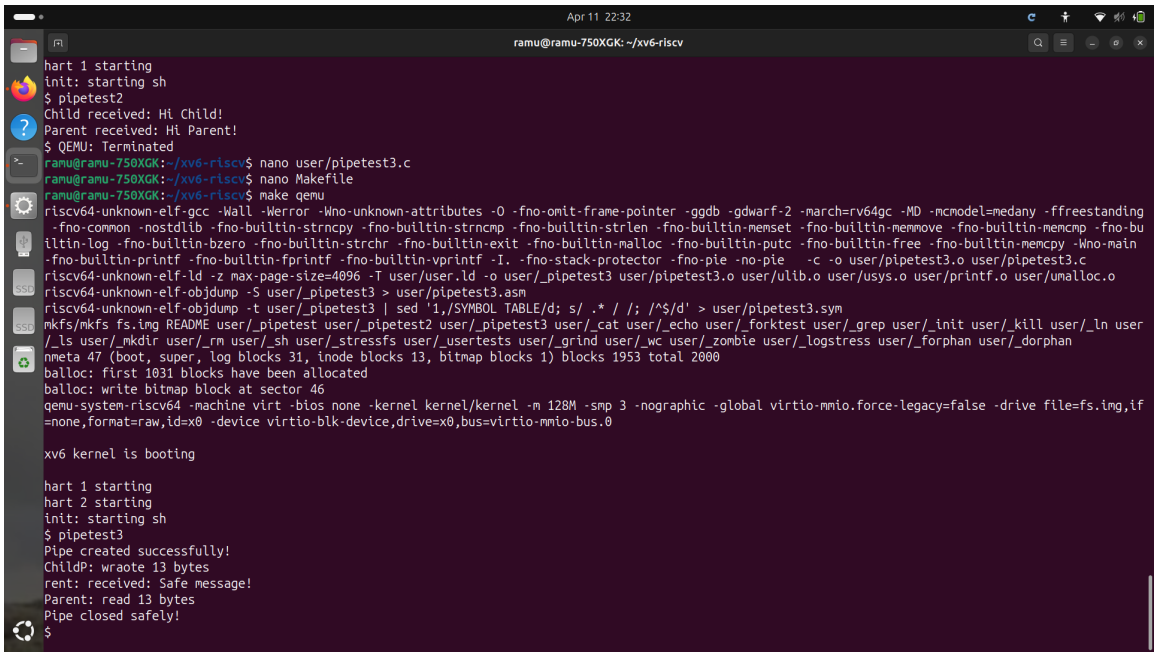
How to run:

```
$ pipetest3
```

Output:

```
Pipe created successfully!
Child: wrote 13 bytes
Parent: received: Safe message!
Parent: read 13 bytes
Pipe closed safely!
```

Screenshot:



How Pipes Work in xv6

1. `pipe(fd)` creates two file descriptors
   - `fd[0]` = read end
   - `fd[1]` = write end
2. `fork()` creates child process
3. Child writes to `fd[1]`
4. Parent reads from `fd[0]`
5. `close()` closes unused ends

Conclusion

Successfully implemented 3 pipe functionalities in xv6 operating system demonstrating IPC using pipes including basic, bidirectional and error-handled pipe communication.